

De Novo Design of κ -Opioid Receptor Antagonists Using a Generative Deep-Learning Framework

Leslie Salas-Estrada, Davide Provasi, Xing Qiu, Husnu Ümit Kaniskan, Xi-Ping Huang, Jeffrey F. DiBerto, João Marcelo Lamim Ribeiro, Jian Jin, Bryan L. Roth, and Marta Filizola*



Cite This: <https://doi.org/10.1021/acs.jcim.3c00651>



Read Online

ACCESS |



Metrics & More

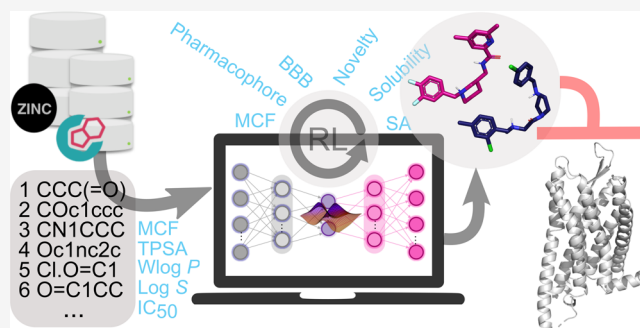


Article Recommendations



Supporting Information

ABSTRACT: Likely effective pharmacological interventions for the treatment of opioid addiction include attempts to attenuate brain reward deficits during periods of abstinence. Pharmacological blockade of the κ -opioid receptor (KOR) has been shown to abolish brain reward deficits in rodents during withdrawal, as well as to reduce the escalation of opioid use in rats with extended access to opioids. Although KOR antagonists represent promising candidates for the treatment of opioid addiction, very few potent selective KOR antagonists are known to date and most of them exhibit significant safety concerns. Here, we used a generative deep-learning framework for the *de novo* design of chemotypes with putative KOR antagonistic activity. Molecules generated by models trained with this framework were prioritized for chemical synthesis based on their predicted optimal interactions with the receptor. Our models and proposed training protocol were experimentally validated by binding and functional assays.



INTRODUCTION

Opioids are widely prescribed medications for the treatment of chronic or severe pain. However, repeated exposure to these drugs can lead to neurological adaptations in the brain that result in addiction.^{1,2} This chronic, relapsing brain disease, also known as opioid use disorder (OUD), currently affects sixteen million individuals worldwide.² It is characterized by compulsive drug seeking and use despite adverse effects, including respiratory depression, which has recently resulted in approximately 1000 deaths every week in the United States alone, according to recent estimates.³ Addiction to these drugs has led to a global opioid overdose epidemic that has been exacerbated in recent years due to the COVID-19 pandemic.^{4–6} Currently, this ongoing crisis is being addressed from different angles, including prevention of exposure and addiction, support for the discovery of new analgesics, improvement of clinical guidelines for pain management, and the development of treatments for OUD.^{7–9}

Suggested priority mechanisms for the development of therapeutics to treat OUDs¹⁰ include pharmacological blockade of the activity of the κ -opioid receptor (KOR), a member of the G protein-coupled receptor (GPCR) superfamily. This approach has shown promise in reducing withdrawal symptoms and modulating other key elements linked to addiction such as reward and stress responses.^{11–19} However, the set of known KOR antagonists is limited, and most of them present limitations, including persistent pharmacodynamics, delayed onset of effects, poor brain penetration, and unwanted

side effects.^{11,20–24} Thus, there is a need to broaden the availability of novel KOR antagonists to improve our understanding of the receptor's function and to develop safer and more effective treatments for opioid addiction.

Recent applications of artificial intelligence (AI)-based tools have demonstrated significant potential in reducing the resources required for traditional drug discovery pipelines, which have estimated cycle times and costs that can range from 10–20 years and 0.8–2.3 billion US dollars, respectively.^{25,26} Specifically, generative deep-learning models have been successfully implemented to solve molecular design and drug development problems.^{27–39} Using architectures composed of neural networks with multiple layers, these models can learn to approximate the probability distribution of the samples used to train them and then create new samples (not present in the input data set) from this learned distribution.

In this study, we used a modified version of the generative tensorial reinforcement learning (GENTRL) architecture⁴⁰ to design KOR antagonists *de novo*. The architecture comprises a conditional variational autoencoder (cVAE) that uses a recurrent neural network (RNN) encoder and a convolutional

Received: April 29, 2023

neural network (CNN) decoder. Recently, this architecture was shown to facilitate the rapid discovery of inhibitors of the discoidin domain receptor 1 from design to *in vivo* verification⁴⁰ by utilizing a two-step protocol consisting of pretraining, followed by reinforcement learning (RL) cycles. Here, we first conducted pretraining on a data set of millions of drug-like compounds in a Simplified Molecular Input Line Entry System (SMILES) format to learn an approximation of the distribution of the high-dimensional chemical space and then on a smaller, specialized data set that included known KOR inhibitors from the ChEMBL 30 database⁴¹ in an attempt to fine-tune the model. The high-dimensional chemical space was mapped by the cVAE into a low-dimensional latent space that encoded the relationships between SMILES and selected molecular properties for use in the generative model to sample compounds within the same learned distribution but beyond the known chemical space. An established reinforcement learning algorithm was then used to bias compound generation toward new chemotypes that were predicted to inhibit KOR's activity. Five molecules were prioritized for experimental testing based on their high ligand–receptor structural interaction fingerprints (SIFt) and shape similarity to the JDTic crystallographic binding pose at KOR.⁴² These molecules were synthesized and experimentally tested using binding and functional assays. All five compounds were able to bind to the receptor, and the two with the highest binding affinity were functionally evaluated and found to exhibit antagonism at KOR.

MATERIALS AND METHODS

Generative Deep-Learning Architecture. The conditional variational autoencoder within the GENTRL's open framework⁴⁰ was used as the basis for our generative modeling strategy. The encoder consisted of a 2-layer RNN with gated recurrent units of hidden size 256, while the decoder consisted of a 128-channel dilated CNN with 7 stacked layers. A 50-dimensional latent space was parameterized using a tensor decomposition with mixtures of 20 Gaussian components at each dimension and tensor trains with a core size of 30. The architecture was modified to correctly encode and decode all entries in the training data sets by extending the vocabulary to include all of the necessary characters and increasing the maximum allowed length of individual entries (see below).

Pretraining Data Sets, Properties, and Evaluation. We used two different data sets to pretrain the generative model. Data set #1 consisted of a curated list of 4.6 million compounds. To obtain this curated data set, we started with the 6.2 million two-dimensional (2D), clean, in-stock, drug-like compounds from the ZINC 20 database⁴³ (as of 03/08/2021) and removed molecules that had atoms besides C, N, S, O, F, Cl, Br, and H, cycles with more than 8 atoms, more than 7 rotatable bonds, and potentially toxic and/or reactive moieties based on medicinal chemistry filters (MCFs) and pan assay interference compounds (PAINS) filters.^{44–46} Next, we partitioned the curated set into three subsets: a pretraining data set (80%), a test set (10%), and a scaffold test set (10%). The scaffold test set consisted of molecules with scaffolds assessed using the Bemis–Murcko method⁴⁷ that were absent in both the pretraining data set and the test set. To achieve this split, we employed the `split_dataset` function of the `prepare_dataset.py` script from the Molecular Sets (MOSES) benchmarking platform (https://github.com/molecularsets/moses/blob/master/scripts/prepare_dataset.py) and followed

a methodology described in the literature.⁴⁸ The scaffold test set was used to evaluate the generative model's ability to produce molecules with novel scaffolds. Data set #2 consisted of 2,056 compounds from the ChEMBL 30 database⁴¹ (as of 01/27/2022) with reported half-maximum inhibitory concentration (IC_{50}) at human KOR (single-protein target ChEMBL ID 237). The data set was divided into two categories: “positive” KOR inhibitors ($IC_{50} < 1 \mu M$) and “negative” KOR inhibitors ($IC_{50} \geq 1 \mu M$), consisting of 445 and 1611 molecules, respectively. Both pretraining data sets were saved in the SMILES format.

Five molecular properties were calculated for all molecules in the pretraining data sets: (1) the topological polar surface area (TPSA), estimated using RDKit's TPSA descriptor based on contributions from N, O, S, and P atoms;^{49,50} (2) lipophilicity as per Wildman–Crippen's $\log P$ ($W \log P$), calculated using RDKit's Mol Log P descriptor with an atom-based approach;^{50,51} (3) solubility as per $\log S$, estimated using Delaney's Estimated SOLubility (ESOL) method;⁵² (4) a binary flag indicating whether the molecule passed (1) or failed (0) MCFs; and (5) a binary flag indicating whether the molecule was a KOR inhibitor (1) or a noninhibitor (0) based on its reported IC_{50} at KOR. This property was only available for molecules in data set #2 and their corresponding entries in data set #1. For molecules without IC_{50} records, the corresponding property was left empty to indicate it as missing.

The model was initially pretrained on data set #1 for up to 25 epochs using an Adam optimizer with a learning rate of 10^{-4} . Each epoch represents one pass over the complete data set in training batches of 50 SMILES.⁵³ The quality of the model and its ability to approximate the distribution learned from data set #1 was assessed using established metrics⁴⁸ and 30,000 valid SMILES sampled after 0, 2, 4, 6, 8, 10, 15, 20, and 25 epochs of pretraining. These metrics included validity, uniqueness, internal diversity, and passing chemical filters. Validity was assessed by verifying successful/failed conversion to RDKit's and OpenEye's molecule objects.^{50,54} Uniqueness was defined as the fraction of valid SMILES that were not duplicated in a generated set of molecules.⁴⁸ Internal diversity was measured as 1 minus the average chemical similarity as per the Tanimoto coefficient (T_c) within the generated batch using Morgan fingerprints.⁴⁸ The same chemical filters used during the construction of pretraining data set #1 were used to evaluate the fraction of molecules passing chemical filters. The generated molecules were also compared to the reference data sets (test set and scaffold test set) using established metrics to assess the quality of generative models.⁴⁸ These metrics included BRICS fragments similarity,⁵⁵ Bemis–Murcko scaffold⁴⁷ similarity, molecular properties (e.g., molecular weight, lipophilicity, synthetic accessibility, and quantitative estimate of drug-likeness), as well as the Fréchet ChemNet Distance (FCD)⁵⁶ and average Tanimoto similarity of Morgan fingerprints to the nearest neighbor in the reference sets.

In the second step of pretraining, the model underwent 300,000 updates using an Adam optimizer with a learning rate of 10^{-4} . An update consists of a pass over one batch of 200 molecules, which included 100 molecules from data set #1, 50 molecules from the “positive” set in data set #2, and 50 molecules from the “negative” set in data set #2.

Reinforcement Learning. Reinforcement learning was performed using the REINFORCE algorithm⁵⁷ within the GENTRL's reinforcement learning framework. The training was carried out for 9000 iterations, using an Adam optimizer

with a learning rate of 10^{-5} and a batch size of 200. An iteration consists of one batch of 200 SMILES generated and scored by the model. The reward function was computed as the product of six reward components, calculated as follows.

Pharmacophoric Similarity (Pharm). The OpenEye Python Toolkits (version 2021.2.0)⁵⁴ was used to generate 3D conformers of SMILES, following the protocol outlined in ref 58. Briefly, the most probable tautomeric form and protonation state at pH 7.4 were determined using the Quacpac Toolkit (version 2.1.3.0),⁵⁹ while 3D structure generation was carried out with the Omega Toolkit (version 4.1.2.0),⁶⁰ which performed stereoisomer enumeration for up to 3 stereocenters and a maximum of 200 conformers. To assess similarity to JDTC's binding mode at KOR (PDB ID 4DJH),⁴² two methods were used: (a) the Ultrafast Shape Recognition with the CREDO Atom Type score⁶¹ (USRCAT score) or (b) the Shaper2⁶² alignment score to cavity-derived pharmacophoric features determined with the IChem toolkit⁶³ (IChem score). To calculate USRCAT scores, the OpenEye-generated 3D conformers were converted into RDKit molecule objects using the Python module available in ref 64 and then ranked according to their USRCAT scores. To calculate IChem scores, chain B of the human inactive JDTC-bound KOR structure (PDB ID 4DJH)⁴² was prepared using the default protocol of the Protein Preparation Wizard in the Schrödinger 2020-4 suite.⁶⁵ The protocol consisted of (i) back mutating residue L135 to isoleucine to match the wild-type sequence; (ii) adding missing side chains and hydrogen atoms; (iii) capping the N- and C-terms of the chain at residues S55 and P347, respectively; (iv) assigning correct bond orders and protonation states for the ligand and the side chains at pH 7.4; and (iv) conducting a short, restrained energy minimization *in vacuo* using the OPLS3e force field.⁶⁶ Then, mol2 files of the prepared KOR and JDTC structures were generated using ChimeraX,^{67,68} and IChem's Volsite tool was used in the ligand-restricted mode to derive pharmacophoric features from the cavity around the ligand, as described in ref 62. The overlap of conformers to the cavity-based pharmacophore was assessed with the Shaper2 tool using default parameters. The Pharm reward component was defined as either the best USRCAT or IChem score among the generated conformers.

Medicinal Chemistry Filters (MCFs). MCFs were evaluated using substructure matching against a library of SMARTS patterns from the MOSES platform.⁴⁸ The MCF reward component was defined with a binary flag that indicated whether a molecule passed (1) or (0) failed the filters.

Blood–Brain Barrier Permeability (BBB). The BBB was estimated using the Brain Or Intestinal Estimated permeation (BOILED-Egg) method, which is a graphical model that predicts brain access based on a molecule's $W \log P$ and TPSA.^{49,69} The BBB reward component was defined with a binary flag that indicated whether a molecule was predicted to be BBB penetrant (1) or not (0).

Solubility (Log S). Log S was estimated using Pat Walters' implementation of the ESOL method,⁵² which is available on GitHub (<https://github.com/PatWalters/solubility>). This method is based on a linear regression model derived from ~2900 solubility measurements that include parameters such as $\log P_{\text{octanol}}$, molecular weight, the proportion of heavy atoms in aromatic systems, and the number of rotatable bonds. The solubility reward component was defined with a binary flag that indicated whether the predicted log S was between -5 and -1 ⁷⁰ (1) or not (0).

Ease of Synthesis (SA). SA was evaluated using the RDKit implementation of Ertl and Schuffenhauer's synthetic accessibility score (SAS),⁷¹ which can be found on GitHub at https://github.com/rdkit/rdkit/tree/master/Contrib/SA_Score. This score ranks molecular complexity based on a library of substructures on a scale of 1 (easiest) to 10 (hardest), where $SAS > 6$ is considered difficult to synthesize.⁴⁹ The SA reward component was defined as the normalized value $(10 - SAS)/9$, where a score of 0 indicates the hardest molecules to synthesize and 1 indicates the easiest.⁷²

Novelty against Known KOR Inhibitors (Novelty). The novelty of the generated molecules was assessed by computing the maximum Tanimoto coefficient ($T_{\text{c}_{\text{max}}}$) between the Morgan topological 2D fingerprints of each generated molecule and those of known KOR antagonists, specifically the 445 compounds with $IC_{50} < 1 \mu\text{M}$ at KOR included in the "positive" subset of data set #2. Morgan topological 2D fingerprints were calculated using RDKit's Chem module and RDKit's DataStructs module was used to compute their Tanimoto similarity. The novelty reward component was defined as $1 - T_{\text{c}_{\text{max}}}$, where a score of 0 indicates that the molecule is not novel and a score of 1 indicates novel molecules.

A reward of 0 was assigned to invalid molecules, as determined by the failed conversion of SMILES to RDKit and/or OpenEye molecule objects,^{50,54} or to molecules for which the generation of 3D conformers was unsuccessful.

Filtering. The generated molecules were scored using the six reward components defined in the reinforcement learning stage and selected for docking if they met the following criteria: a USRCAT score > 0.25 or an IChem score > 0.4 , passing MCFs, likely BBB permeability, $-5 \leq \log S \leq -1$, $SAS < 6$, and $T_{\text{c}_{\text{max}}} < 0.5$.

Molecular Docking. The molecules were prepared for docking with the LigPrep utility included in the Schrödinger 2020-4 suite.⁶⁵ The ionization state and tautomeric forms determined by the pharmacophoric similarity reward component were preserved during ligand preparation. The human inactive JDTC-bound KOR structure (PDB ID 4DJH)⁴² was prepared for docking using the default protocol of the Protein Preparation Wizard in the Schrödinger 2020-4 suite,⁶⁵ as described above. A cubic docking grid with an edge of 26 Å was generated on the center of mass of JDTC using Schrödinger's Receptor Grid Generation utility.⁶⁵ The molecules were docked using Glide with the Extra Precision (XP) scoring function,⁶⁵ and JDTC was docked using the same procedure as a control.

Structural Interaction Fingerprints (SIFs). An in-house Python script was used to calculate the following types of protein–ligand interactions between the docked molecules and KOR residues: (1) carbon–carbon interactions within 4.5 Å (Apolar), (2) face-to-face aromatic interactions within 4 Å (Aro_F2F), (3) edge-to-face aromatic interactions within 4 Å (Aro_E2F), (4) hydrogen bonds with a protein residue as the donor (Hbond_ProD), (5) hydrogen bonds with a protein residue as the acceptor (Hbond_ProA), (6) electrostatic interactions (within 4 Å) with a protein residue positively charged (Elec_ProP), and (7) electrostatic interactions (within 4 Å) with a protein residue negatively charged (Elec_ProN). The similarity of the poses of the docked molecules to JDTC's pose in the JDTC-bound crystal structure (PDB ID 4DJH)⁴² was assessed by Tanimoto coefficients of their respective SIFs using RDKit.⁵⁰

Chemical Synthesis. Common materials or reagents were purchased from commercial sources and used without further purification. High-performance liquid chromatography (HPLC) spectra were acquired using an Agilent 1200 Series system with a DAD detector for all of the intermediates and final products below. Chromatography was performed on a 2.1 × 150 mm Zorbax 300SB-C18 5 μ m column with water containing 0.1% formic acid as solvent A and acetonitrile containing 0.1% formic acid as solvent B at a flow rate of 0.4 ml/min. The gradient program was as follows: 1% B (0–1 min), 1–99% B (1–4 min), and 99% B (4–8 min). High-resolution mass spectra (HRMS) data were acquired in positive-ion mode using an Agilent G1969A API-TOF with an electrospray ionization (ESI) source. Ultraperformance liquid chromatography (UPLC) spectra for compounds were acquired using a Waters Acquity I-Class UPLC system with a photodiode array (PDA) detector. Chromatography was performed on a 2.1 Å ~30 mm Acquity UPLC BEH C18 1.7 μ m column with water containing 3% acetonitrile and 0.1% formic acid as solvent A and acetonitrile containing 0.1% formic acid as solvent B at a flow rate of 0.8 mL/min. The gradient program was as follows: 1–99% B (1–1.5 min) and 99–1% B (1.5–2.5 min). Preparative HPLC was performed on an Agilent Prep 1200 series with a UV detector set to 220 or 254 nm. Samples were injected onto a Phenomenex Luna 250 × 30 mm, 5 μ m, C₁₈ column at room temperature. The flow rate was 40 ml/min. A linear gradient was used with 10% of acetonitrile in H₂O (with 0.1% TFA) (B) to 100% of acetonitrile (A). Nuclear magnetic resonance (NMR) spectra were acquired on a Bruker DRX-400 spectrometer with 400 MHz for protons (¹H NMR) or a Bruker Avance-III 800 MHz spectrometer with 800 MHz for protons (¹H NMR); chemical shifts are reported in (δ). HPLC was used to establish the purity of target compounds. All final compounds had >96% purity using the HPLC methods described above. The chemical synthesis of compounds (S)-N-((1-(3,4-difluorobenzyl)piperidin-3-yl)methyl)-4,6-dimethylpicolinamide (FJRL-14), 2-((2-chloro-4-methylbenzyl)(methyl)amino)-1-(4-(4-chlorobenzyl)-1,4-diazepan-1-yl)ethan-1-one (FJRL-20), (S)-N-((1-(3,4-difluorobenzyl)piperidin-3-yl)methyl)-4,6-dimethylpicolinamide (FJRL-29), 1-((6-methylpyridin-2-yl)methyl)-N-(2-(pyridin-3-yl)ethyl)azepane-4-carboxamide (FJRL-36), and N-((1-(4-isopropylbenzyl)piperidin-4-yl)methyl)-4-methylbenzamide (FJRL-91) is described in detail in the [Supporting Information](#).

Competition Binding Assays. Radioligand binding assays were carried out using membranes prepared from HEK293 cells transiently transfected with human KOR. Samples were mixed with KOR membranes and ³H-U69593 (~1 nM) in 96-well plates. After incubation of 1 h at room temperature, the radioligand bound to KOR membranes was filtered and separated from the free radioligand. Radioactivity was measured with a MicroBeta counter. Results were compared to reference Salvinorin A (Sal A) (0–100%) and fitted to the built-in one-site function for competitive binding assays in GraphPad Prism (v9.5).⁷³

Cell-Based Functional Assays. G_i GloSensor cAMP assays were carried out in transiently transfected HEK293 T cells. Overnight transfected cells (KOR + GloSensor) were plated in poly-L-Lys-coated 384-well white plates in DMEM supplemented with 1% dialyzed FBS for 6 h. After the medium was removed, cells were stimulated with samples premixed with a luciferase substrate for 10 min at room temperature,

followed by the addition of isoproterenol (100 nM) to activate endogenous β_2 -adrenergic receptor-stimulated G_s signaling. The plate was then read after 20 min by a luminescence reader. For antagonist activity, 10 nM Sal A was added at 10 min after samples for additional 10 min incubation before isoproterenol. Results (relative luminescence units) were normalized to basal (1.0 fold of basal) and were analyzed with the built-in four-parameter concentration-response function in GraphPad Prism.⁷³ Reduction in luminescence is proportional to G_i activation.

RESULTS

Distribution Learning and De Novo Compound Generation. Figure 1 shows a schematic representation of

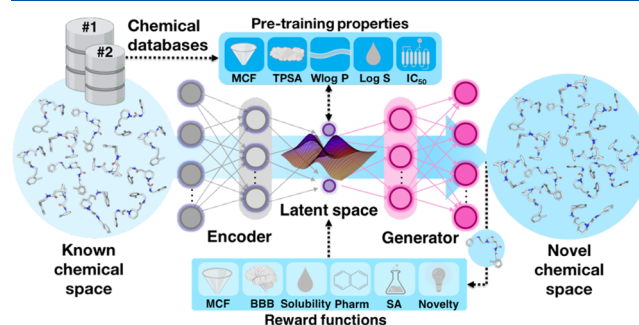


Figure 1. Scheme of the generative deep-learning strategy used for the *de novo* design of KOR antagonists. MCF: medicinal chemistry filters; TPSA: topological polar surface area; Wlog P: Wildman–Crippen’s log P; Log S: logarithm of solubility; BBB: blood–brain barrier; Pharm: pharmacophore similarity; and SA: ease of synthesis. Some elements of this figure were created with BioRender (BioRender.com).

the generative deep-learning strategy that we used to design KOR ligands with antagonistic activity, leveraging the structural and physicochemical information of the receptor, its ligands, and large chemical libraries of drug-like molecules.

Specifically, we used a modified version of the GENTRL architecture⁴⁰ and pretrained a model using data derived from 4.6 million molecules from the ZINC 20 database⁴³ (data set #1). These molecules were characterized by molecular properties such as MCFs,⁴⁸ TPSA, Wlog P, log S,^{52,70} and reported KOR inhibition (IC₅₀) (see [Materials and Methods](#) for further details). We evaluated the performance of this model by conducting pretraining for an increasing number of epochs. Each epoch involved the model passing over the entire data set #1 once. We assessed the model’s ability to generate chemically valid, novel, and diverse drug-like compounds and to approximate the distribution learned from data set #1 using established metrics⁴⁸ (see [Materials and Methods](#) for details). [Figure S1](#) illustrates the model’s performance in generating 30,000 valid SMILES as the number of pretraining epochs increases. The highest validity values were achieved after 20 epochs, and we selected the resulting model for fine-tuning via a second round of pretraining using data set #2. This data set consisted of 300,000 batches, each containing 100 molecules from data set #1, 50 KOR inhibitors and 50 KOR noninhibitors (see [Materials and Methods](#) for details). The performance of the model in producing 30,000 valid SMILES after the second round of pretraining is also depicted in [Figure S1](#), whereas [Tables S1 and S2](#) provide estimates of the model’s performance on reference data sets after pretraining on data set

#1 or #2, respectively. Notably, 99–100% of molecules generated by the model are novel, meaning that they are not present in the training set. The parameters obtained from the model after 20 epochs were saved and used as the starting point for a reinforcement learning step.

Reinforcement Learning for *De Novo* Generation of KOR Antagonists. Two separate models were trained using a multicomponent reward function computed as the product of the reward components described in the [Materials and Methods](#) section. These components included an estimate of ligand-based or cavity-based pharmacophoric similarity to the ligand binding mode in the JDTic-bound KOR structure,⁴² expressed as a USRCAT score or IChem score, respectively. The performance of the two models in producing 30,000 valid molecules was monitored during reinforcement learning by a set of indicators, including the fraction of molecules with a combined reward greater than 0.1 (Figure 2), as well as the

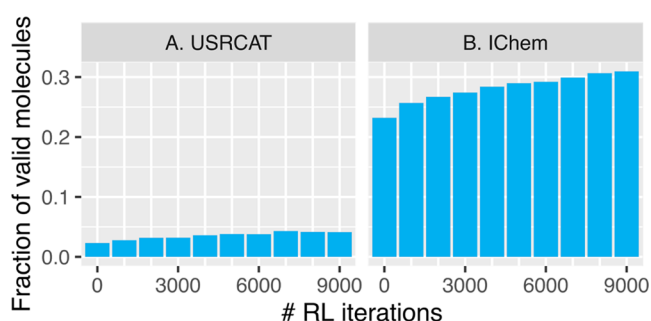


Figure 2. Performance of the generative deep-learning models in producing molecules highly scored by a multicomponent reward function. (A) Model trained using ligand-based pharmacophoric similarity assessed by USRCAT scores. (B) Model trained using cavity-based pharmacophoric similarity assessed by IChem scores. Each reinforcement learning iteration consists of the model generating and scoring 200 compounds in the SMILES format.

validity of chemical structures, their internal diversity, uniqueness, and passing of chemistry filters (Figure S2). The training improved learning in both models up to 9000 reinforcement learning iterations, but incorporating cavity-based information produced a larger fraction of valid molecules.

Compound Prioritization for Chemical Synthesis and Experimental Testing. Figure 3 illustrates the strategy used to prioritize molecules for chemical synthesis and experimental testing. First, the two trained models were used to generate sets of 500,000 valid SMILES each after 9000 reinforcement learning iterations, resulting in a total of 1 million SMILES. Compounds that met the desired reward values for ligand-based or cavity-based pharmacophore similarity (USRCAT > 0.25 or IChem > 0.4, respectively), novelty ($T_{c_{max}} < 0.5$), ease of synthesis ($SAS < 6$), solubility ($-5 \leq \log S \leq -1$), and likelihood to penetrate the BBB while passing MCFs were selected for molecular docking. In total, 2545 compounds satisfied these criteria, with 1116 and 934 molecules derived from the model using USRCAT and IChem scores, respectively. The best 3D conformers of these molecules, as determined by the pharmacophoric similarity reward component, were prepared for docking at KOR (PDB ID 4DJH)⁴² in the same ionization state and tautomeric form. The docked compounds using Glide XP⁶⁵ and JDTic as the docking control were ranked by their similarity to JDTic's interactions with

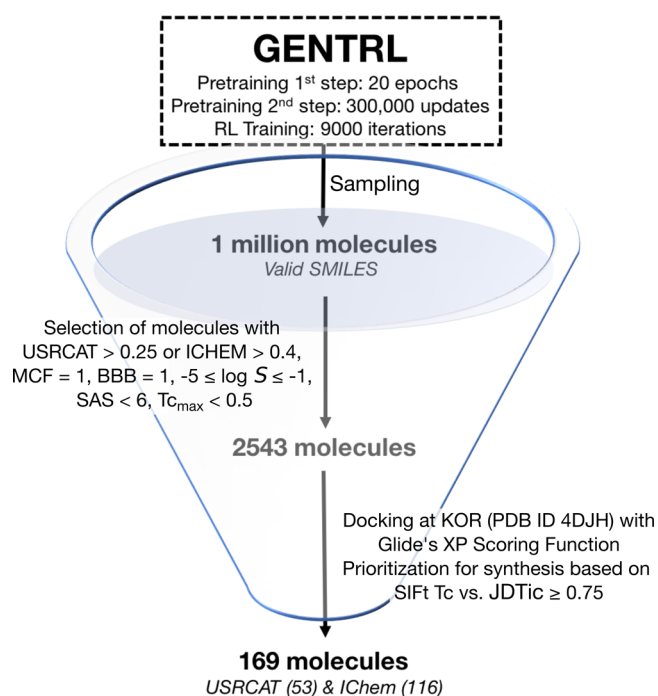


Figure 3. Prioritization pipeline for chemical synthesis and experimental testing of compounds generated by models trained with reinforcement learning. Two sets of 500,000 compounds each were generated from models using USRCAT and IChem scores.

KOR. Table S3 shows the 2D structures of the 169 molecules exhibiting the highest similarity to JDTic's interactions with KOR, as assessed by SIFT Tc values larger than 0.75. Notably, more molecules generated by the model using IChem scores had ligand–receptor interactions closer to JDTic's than those generated by the model using USRCAT scores (116 vs 53 compounds with SIFT Tc ≥ 0.75).

Five top-scoring docked compounds (FJRL-14, FJRL-20, FJRL-29, FJRL-36, and FJRL-91) based on SIFT Tc values were selected for chemical synthesis and experimental validation based on the ready availability of chemical materials. Details of the chemical synthesis for these compounds are reported in the [Materials and Methods](#) section and [Supporting Information](#). As expected, based on their similarity in binding mode with JDTic, all five compounds were predicted to have salt-bridge or hydrogen-bond interactions with KOR residue D138^{3,32} in transmembrane (TM) 3 (Figure S3). Superscript numbers here and throughout refer to Ballesteros–Weinstein's generic numbering scheme for GPCRs.^{74,75} Additionally, all compounds shared hydrophobic interactions with residues in TM2 (V108^{2,53}), TM3 (V134^{3,28}, I135^{3,29}, D138^{3,32}, Y139^{3,33}, M142^{3,36}), TM6 (W287^{6,48}, I290^{6,51}, I294^{6,55}), and TM7 (I316^{7,39} and Y320^{7,43}) (Figure S4).

Experimental Validation of Synthesized Compounds. Competition binding assays were performed for compounds FJRL-14, FJRL-20, FJRL-29, FJRL-36, and FJRL-91 according to the protocols described in the [Materials and Methods](#) section. All five tested compounds were found to bind to the receptor at high concentrations with K_i values of 6.46 μM (pK_i of 5.19 $\pm$ 0.06) and 7.59 μM (pK_i of 5.12 $\pm$ 0.06) for FJRL-14 and FJRL-20, and more than 10 μM for FJRL-29, FJRL-36, and FJRL-91, respectively, compared to JDTic and Salvinorin A, which were used as positive controls (pK_i = 8.45 $\pm$ 0.10, K_i = 3.55 nM, and pK_i = 8.30 $\pm$ 0.13, K_i = 5.01 nM, respectively).

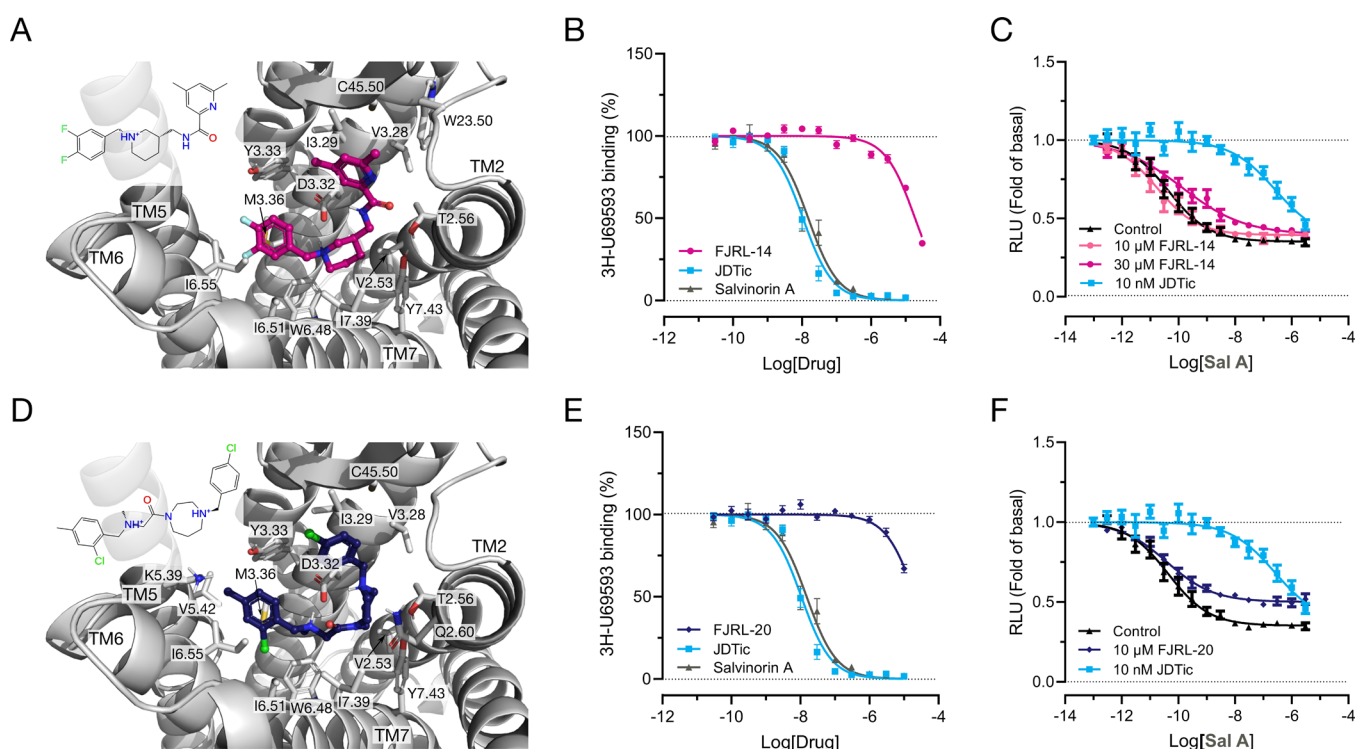


Figure 4. *De novo*-generated KOR antagonists FJRL-14 and FJRL-20. (A, D) Docked compounds at the JDTic-KOR's crystal structure (PDB ID 4DJH) in both 2D (lines) and 3D (sticks) formats with interacting residues as per structural interaction fingerprints shown as sticks. (B, E) Competition binding assays using KOR membranes and 1 nM ^3H -U69593 for FJRL-14 and FJRL-20, respectively. (C, F) KOR-mediated G_i inhibition of cAMP production as assessed in transiently transfected HEK293 T cells using the GloSensor cAMP assay (Promega) for FJRL-14 and FJRL-20, respectively.

Figure 4 shows the two compounds with the highest binding affinity for KOR, specifically, FJRL-14 and FJRL-20, docked at the KOR binding pocket (Figure 4A,D, respectively), as well as their results from competition binding assays using KOR membranes and 1 nM ^3H -U69593 (Figure 4B,E, respectively), and Sal A-mediated G_i inhibition of cAMP production as assessed in transiently transfected HEK293 T cells (Figure 4C,F, respectively). FJRL-14 and FJRL-20 showed no agonist activity (results not shown) but weak antagonist activity against Sal A-mediated G_i activation. Details of the experimental protocols are provided in the [Materials and Methods](#) section.

As also shown in Figure S3, the predicted binding mode of FJRL-14 and FJRL-20 at KOR is very similar to JDTic's, with the ligands sharing most ligand–receptor interactions (Figure S4). However, despite this high level of similarity, at least one of the two compounds is missing interactions with KOR that are present for JDTic (e.g., Q115^{2,60}, G319^{7,42}) and are not replaced by an equivalent number of new interactions. This reduced number of interactions between the *de novo* compounds and the receptor is likely responsible for the observed lower binding affinities as per radioligand binding assays and weaker antagonistic activities as determined by Schild analysis of competitive antagonism. Confirmation of the novelty of FJRL-14 and FJRL-20, in comparison to the reported 337 opioid ligands in the IUPHAR/BPS Guide to Pharmacology database,⁷⁶ was achieved by calculating Tc_{max} values below 0.5 and examining the closest 2D structures. A chemical similarity search between FJRL-14 and FJRL-20 with the entire ChEMBL 30 database⁴¹ (~2.2 million compounds)

resulted in Tc_{max} values of 0.67 and 0.70, respectively, providing further evidence of their chemical novelty.

DISCUSSION

In recent years, AI-based tools have shown great potential for enhancing drug research and development, particularly through the use of generative deep-learning architectures. However, the validation of AI-assisted novel molecule identification remains a crucial challenge, especially for central nervous system diseases, often due to a shortage of high-quality data to train reproducible ML models, difficulties in interpreting data derived from machine learning (ML) models, and the challenging need to integrate data from various sources. Despite these limitations, our work demonstrates that generative deep-learning models offer a promising solution to accelerate the early lead discovery process for members of the GPCR superfamily.

In our study, we used a generative modeling architecture that had been experimentally shown to generate potent leads in just 21 days from conception to *in vivo* validation. However, we had to modify it to adapt it to our specific problem and to compensate for the lack of accessibility to proprietary information, including the code for the reinforcement learning reward functions. Thus, we utilized publicly available databases of chemical compounds from which to derive drug-like properties, as well as structural and functional information about KOR, to establish and validate a training protocol for generating novel KOR antagonists. Our work demonstrates that this revised protocol and algorithm enables the exploration of uncharted regions of the chemical space and provides a viable alternative to biased chemical library screening.

While the compounds we have been able to synthesize and test so far exhibited weak antagonistic activity, our results confirm that these tools can effectively design new compounds with desired pharmacological profiles *in vitro*. However, further studies are needed to verify whether these molecules also have desirable pharmacological profiles *in vivo*, such as BBB permeability, which we included as a reward function. Therefore, we are currently working on incorporating potency into our generative architecture to enhance the possibility of obtaining potent compounds that can be candidates for animal testing. Additionally, we are conducting structure–activity relationship studies to enhance the affinity of the confirmed chemical scaffolds thus far.

One area where we see the potential for further improvement is increasing the yield of valid molecules, although this limitation could be circumvented by increasing the number of molecules being generated. The current low yield of valid molecules is likely due to using an RNN for the encoder, which becomes harder to train with longer input sequences. In addition to using different encoders, testing different physicochemical properties and structural evaluations for training could also improve the yield of valid molecules.

Additionally, we found that using reward functions derived from cavity-based information allowed us to generate more molecules with ligand–receptor interactions closer to JDTic's than reward functions derived from ligand-based molecular shape features. Future work is required to investigate different hybrid ligand- and structure-based training schemes.

It is worth noting that our protocol can be generalized to other GPCRs with available three-dimensional structures or molecular models by updating the data sets used for training with sets that are relevant to the GPCR under study. As new structural and functional information on GPCRs becomes available, we expect that the performance of AI-driven molecular design models will continue to improve and have a further impact on reducing the times and costs associated with drug discovery for the treatment of various conditions, including opioid addiction.

CONCLUSIONS

While there are challenges associated with applying AI/ML technology to drug discovery, our work highlights the potential benefits of using generative deep-learning models to accelerate the early lead discovery process for GPCRs. Our study demonstrates that AI-driven molecular design can effectively generate novel compounds with desired pharmacological profiles *in vitro*, and we anticipate that continued improvements in AI-driven molecular design models will lead to further advancements in drug discovery for a range of diseases.

ASSOCIATED CONTENT

Data Availability Statement

GENTRL and MOSES were obtained from their respective GitHub repositories (<https://github.com/insilicomedicine/GENTRL> and <https://github.com/molecularsets/moses>). Chemical libraries were downloaded from the ZINC 20 database (<https://zinc20.docking.org/>) and the ChEMBL 30 database (<https://www.ebi.ac.uk/chembl/>). PDB files were downloaded from the RCSB Protein Data Bank (<https://www.rcsb.org>). Glide XP within the Schrödinger 2020-4 suite was used for docking. SIFT analysis was carried out using an in-house Python script which can be made available upon request. PyMOL(TM) 2.5.2 was used for molecular visualization

(<https://pymol.org/2/>). GraphPad Prism 9.5 was used for the analysis of binding and functional assays. The data sets used as input, generated, or analyzed in this study, are available in a Simplified Molecular Input Line Entry System (SMILES) format on Zenodo (DOI: 10.5281/zenodo.7877934)

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00651>.

Details of chemical synthesis of selected compounds; performance of the generative deep-learning model using reference data sets after pretraining on data set #1 (Table S1); performance of the generative deep-learning model using reference data sets after pretraining on data set #2 (Table S2); list of 169 compounds prioritized for chemical synthesis (Table S3); performance of the generative deep-learning model in producing chemically valid, novel, and diverse molecules during the pretraining steps (Figure S1); performance of the generative deep-learning model in producing chemically valid, novel, and diverse molecules during the reinforcement learning step (Figure S2); compounds prioritized for chemical synthesis docked at KOR and compared to JDTic's binding mode (Figure S3); and protein–ligand interactions of compounds prioritized for chemical synthesis docked at KOR and compared to JDTic's (Figure S4) (PDF)

AUTHOR INFORMATION

Corresponding Author

Marta Filizola – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States; orcid.org/0000-0002-4382-8276; Email: marta.filizola@mssm.edu

Authors

Leslie Salas-Estrada – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States

Davide Provasi – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States; orcid.org/0000-0002-2868-303X

Xing Qiu – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States

Husnu Ümit Kaniskan – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States; orcid.org/0000-0001-5327-832X

Xi-Ping Huang – National Institute of Mental Health, Psychoactive Drug Screening Program, Department of Pharmacology, University of North Carolina School of Medicine, Chapel Hill, North Carolina 27599, United States

Jeffrey F. DiBerto – National Institute of Mental Health, Psychoactive Drug Screening Program, Department of Pharmacology, University of North Carolina School of Medicine, Chapel Hill, North Carolina 27599, United States

João Marcelo Lamim Ribeiro – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States

Jian Jin – Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, New York

10029, United States; Mount Sinai Center for Therapeutics Discovery, Departments of Oncological Sciences and Neuroscience, Tisch Cancer Institute, Icahn School of Medicine at Mount Sinai, New York, New York 10029, United States; orcid.org/0000-0002-2387-3862

Bryan L. Roth – National Institute of Mental Health, Psychoactive Drug Screening Program, Department of Pharmacology, University of North Carolina School of Medicine, Chapel Hill, North Carolina 27599, United States; Division of Chemical Biology and Medicinal Chemistry, University of North Carolina at Chapel Hill Eshelman School of Pharmacy, Chapel Hill, North Carolina 27599, United States; orcid.org/0000-0002-0561-6520

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jcim.3c00651>

Author Contributions

M.F., D.P., L.S.E., and J.M.L.R. conceived and planned the work. L.S.E., D.P., and J.M.L.R. implemented the computational tool. L.S.E. and D.P. analyzed the data. L.S.E. curated the datasets and performed the computations. X.Q. synthesized the prioritized compounds under the supervision of H.U.K. and J.J. X.P.H. and J.F.D. conducted the *in vitro* experiments and analyzed the results under B.L.R.'s supervision. M.F. supervised the project. L.S.E. and M.F. wrote the initial draft of the manuscript with contributions from all authors.

Notes

The authors declare the following competing financial interest(s): J.J. is a cofounder and equity shareholder in Cullgen, Inc., a scientific cofounder and scientific advisory board member of Onsero Therapeutics, Inc., and a consultant for Cullgen, Inc., EpiCypher, Inc., Accent Therapeutics, Inc., and Tavotek Biotherapeutics, Inc. The Jin and Filizola laboratories received research funds from Celgene Corporation. The Jin laboratory also received funds from Levo Therapeutics, Inc., Cullgen, Inc. and Cullinan Oncology, Inc.

ACKNOWLEDGMENTS

This work was funded by the National Institutes of Health grant DA045473. Computations were supported in part through the computational resources and staff expertise provided by Scientific Computing at the Icahn School of Medicine at Mount Sinai, the Clinical and Translational Science Awards (CTSA) grant UL1TR004419 from the National Center for Advancing Translational Sciences, and the Office of Research Infrastructure of the National Institutes of Health under award number S10OD026880. The work utilized the NMR Spectrometer Systems at Mount Sinai acquired with funding from the NIH's Shared Instrumentation Grants 1S10OD025132 and 1S10OD028504. Radioligand binding and GloSensor cAMP assays were performed under the National Institute of Mental Health's Psychoactive Drug Screening Program (NIMH PDSP), contract HHSN-271-2018-00023-C. The NIMH PDSP is directed by Bryan L Roth, MD, PhD at the University of North Carolina at Chapel Hill and Project Officer Jamie Driscoll at NIMH, Bethesda, MD. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

REFERENCES

- (1) Zöllner, C.; Stein, C. Opioids. *Handb. Exp. Pharmacol.* **2006**, *177*, 31–63.
- (2) Azadfrat, M. H.; M R; Leaming, J. M. Opioid Addiction StatPearls 2022.
- (3) U.S. Overdose Deaths in 2021 Increased Half as Much as in 2020 - But Are Still Up 15%. CDC/NCHS, Ed., 2022.
- (4) Mattson, C. L.; Tanz, L. J.; Quinn, K.; Kariisa, M.; Patel, P.; Davis, N. L. Trends and Geographic Patterns in Drug and Synthetic Opioid Overdose Deaths - United States, 2013-2019. *MMWR Morb. Mortal. Wkly Rep.* **2021**, *70*, 202–207.
- (5) UNODC. *World Drug Report 2021*. Ed., 2021.
- (6) CDC. *Emergency Preparedness and Response. Increase in Fatal Drug Overdoses Across the United States Driven by Synthetic Opioids Before and During the COVID-19 Pandemic*. Ed., 2020.
- (7) FDA. *FDA'S 2018 Strategic Policy Roadmap. Healthy Innovation, Safer Families*, Ed., 2018.
- (8) Guy, G. P.; Zhang, K.; Bohm, M. K.; Losby, J.; Lewis, B.; Young, R.; Murphy, L. B.; Dowell, D. Vital Signs: Changes in Opioid Prescribing in the United States, 2006-2015. *MMWR Morb. Mortal. Wkly Rep.* **2017**, *66*, 697–704.
- (9) NIH. *NIH HEAL Initiative Research Plan*. Ed., 2023.
- (10) Rasmussen, K.; White, D. A.; Aciri, J. B. NIDA's medication development priorities in response to the Opioid Crisis: ten most wanted. *Neuropsychopharmacology* **2019**, *44*, 657–659.
- (11) Carlezon, W. A., Jr; Krystal, A. D. Kappa-Opioid Antagonists for Psychiatric Disorders: From Bench to Clinical Trials. *Depression Anxiety* **2016**, *33*, 895–906.
- (12) Schlosburg, J. E.; Whitfield, T. W., Jr; Park, P. E.; Crawford, E. F.; George, O.; Vendruscolo, L. F.; Koob, G. F. Long-term antagonism of kappa opioid receptors prevents escalation of and increased motivation for heroin intake. *J. Neurosci.* **2013**, *33*, 19384–19392.
- (13) McCann, D. J. Potential of buprenorphine/naltrexone in treating polydrug addiction and co-occurring psychiatric disorders. *Clin. Pharmacol. Ther.* **2008**, *83*, 627–630.
- (14) Drapier, D.; Bentue-Ferrer, D.; Laviolle, B.; Millet, B.; Allain, H.; Bourin, M.; Reymann, J. M. Effects of acute fluoxetine, paroxetine and desipramine on rats tested on the elevated plus-maze. *Behav. Brain Res.* **2007**, *176*, 202–209.
- (15) Robinson, S. E. Buprenorphine-containing treatments: place in the management of opioid addiction. *CNS Drugs* **2006**, *20*, 697–712.
- (16) Bagdy, G.; Graf, M.; Anheuer, Z. E.; Modos, E. A.; Kantor, S. Anxiety-like effects induced by acute fluoxetine, sertraline or m-CPP treatment are reversed by pretreatment with the 5-HT_{2C} receptor antagonist SB-242084 but not the 5-HT_{1A} receptor antagonist WAY-100635. *Int. J. Neuropsychopharmacol.* **2001**, *4*, 399–408.
- (17) Xi, Z. X.; Fuller, S. A.; Stein, E. A. Dopamine release in the nucleus accumbens during heroin self-administration is modulated by kappa opioid receptors: An *in vivo* fast-cyclic voltammetry study. *J. Pharmacol. Exp. Ther.* **1998**, *284*, 151–161.
- (18) Knoll, A. T.; Meloni, E. G.; Thomas, J. B.; Carroll, F. I.; Carlezon, W. A., Jr. Anxiolytic-like effects of kappa-opioid receptor antagonists in models of unlearned and learned fear in rats. *J. Pharmacol. Exp. Ther.* **2007**, *323*, 838–845.
- (19) Beardsley, P. M.; Howard, J. L.; Shelton, K. L.; Carroll, F. I. Differential effects of the novel kappa opioid receptor antagonist, JDTC, on reinstatement of cocaine-seeking induced by footshock stressors vs cocaine primes and its antidepressant-like effects in rats. *Psychopharmacology* **2005**, *183*, 118–126.
- (20) Spanagel, R.; Shippenberg, T. S. Modulation of morphine-induced sensitization by endogenous kappa opioid systems in the rat. *Neurosci. Lett.* **1993**, *153*, 232–236.
- (21) Melief, E. J.; Miyatake, M.; Carroll, F. I.; Beguin, C.; Carlezon, W. A., Jr; Cohen, B. M.; Grimwood, S.; Mitch, C. H.; Rorick-Kehn, L.; Chavkin, C. Duration of action of a broad range of selective kappa-opioid receptor antagonists is positively correlated with c-Jun N-terminal kinase-1 activation. *Mol. Pharmacol.* **2011**, *80*, 920–929.

- (22) Potter, D. N.; Damez-Werno, D.; Carlezon, W. A., Jr; Cohen, B. M.; Chartoff, E. H. Repeated exposure to the kappa-opioid receptor agonist salvinorin A modulates extracellular signal-regulated kinase and reward sensitivity. *Biol. Psychiatry* **2011**, *70*, 744–753.
- (23) Van't Veer, A.; Yano, J. M.; Carroll, F. I.; Cohen, B. M.; Carlezon, W. A., Jr. Corticotropin-releasing factor (CRF)-induced disruption of attention in rats is blocked by the kappa-opioid receptor antagonist JDTC. *Neuropsychopharmacology* **2012**, *37*, 2809–2816.
- (24) Urbano, M.; Guerrero, M.; Rosen, H.; Roberts, E. Antagonists of the kappa opioid receptor. *Bioorg. Med. Chem. Lett.* **2014**, *24*, 2021–2032.
- (25) Wouters, O. J.; McKee, M.; Luyten, J. Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009–2018. *JAMA* **2020**, *323*, 844–853.
- (26) Austin, D. H. T. *Research and Development in the Pharmaceutical Industry*, Congressional Budget Office, 2021.
- (27) Sanchez-Lengeling, B.; Aspuru-Guzik, A. Inverse molecular design using machine learning: Generative models for matter engineering. *Science* **2018**, *361*, 360–365.
- (28) Kadurin, A.; Aliper, A.; Kazennov, A.; Mamoshina, P.; Vanhaelen, Q.; Khrabrov, K.; Zhavoronkov, A. The cornucopia of meaningful leads: Applying deep adversarial autoencoders for new molecule development in oncology. *Oncotarget* **2017**, *8*, 10883–10890.
- (29) Kadurin, A.; Nikolenko, S.; Khrabrov, K.; Aliper, A.; Zhavoronkov, A. druGAN: An Advanced Generative Adversarial Autoencoder Model for de Novo Generation of New Molecules with Desired Molecular Properties in Silico. *Mol. Pharmacol.* **2017**, *14*, 3098–3104.
- (30) Putin, E.; Asadulaev, A.; Vanhaelen, Q.; Ivanenkov, Y.; Aladinskaya, A. V.; Aliper, A.; Zhavoronkov, A. Adversarial Threshold Neural Computer for Molecular de Novo Design. *Mol. Pharmacol.* **2018**, *15*, 4386–4397.
- (31) Harel, S.; Radinsky, K. Prototype-Based Compound Discovery Using Deep Generative Models. *Mol. Pharmacol.* **2018**, *15*, 4406–4416.
- (32) Polykovskiy, D.; Zhebrak, A.; Vetrov, D.; Ivanenkov, Y.; Aladinskiy, V.; Mamoshina, P.; Bozdaganyan, M.; Aliper, A.; Zhavoronkov, A.; Kadurin, A. Entangled Conditional Adversarial Autoencoder for de Novo Drug Discovery. *Mol. Pharmacol.* **2018**, *15*, 4398–4405.
- (33) Segler, M. H. S.; Preuss, M.; Waller, M. P. Planning chemical syntheses with deep neural networks and symbolic AI. *Nature* **2018**, *555*, 604–610.
- (34) Merk, D.; Friedrich, L.; Grisoni, F.; Schneider, G. De Novo Design of Bioactive Small Molecules by Artificial Intelligence. *Mol. Inf.* **2018**, *37*, No. 1700153.
- (35) Merk, D.; Grisoni, F.; Friedrich, L.; Gelzinyte, E.; Schneider, G. Computer-Assisted Discovery of Retinoid X Receptor Modulating Natural Products and Isofunctional Mimetics. *J. Med. Chem.* **2018**, *61*, 5442–5447.
- (36) Putin, E.; Asadulaev, A.; Ivanenkov, Y.; Aladinskiy, V.; Sanchez-Lengeling, B.; Aspuru-Guzik, A.; Zhavoronkov, A. Reinforced Adversarial Neural Computer for de Novo Molecular Design. *J. Chem. Inf. Model.* **2018**, *58*, 1194–1204.
- (37) Tong, X.; Liu, X.; Tan, X.; Li, X.; Jiang, J.; Xiong, Z.; Xu, T.; Jiang, H.; Qiao, N.; Zheng, M. Generative Models for De Novo Drug Design. *J. Med. Chem.* **2021**, *64*, 14011–14027.
- (38) Bilodeau, C.; Jin, W.; Jaakkola, T.; Barzilay, R.; Jensen, K. F. Generative models for molecular discovery: Recent advances and challenges. *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **2022**, *12*, No. e1608.
- (39) Özcelik, R.; van Tilborg, D.; Jiménez-Luna, J.; Grisoni, F. Structure-based drug discovery with deep learning. *ChemBioChem* **2022**, No. e202200776.
- (40) Zhavoronkov, A.; Ivanenkov, Y. A.; Aliper, A.; Veselov, M. S.; Aladinskiy, V. A.; Aladinskaya, A. V.; Terentiev, V. A.; Polykovskiy, D. A.; Kuznetsov, M. D.; Asadulaev, A.; Volkov, Y.; Zholus, A.; Shayakhmetov, R. R.; Zhebrak, A.; Minaeva, L. I.; Zagribelnyy, B. A.; Lee, L. H.; Soll, R.; Madge, D.; Xing, L.; Guo, T.; Aspuru-Guzik, A. Deep learning enables rapid identification of potent DDR1 kinase inhibitors. *Nat. Biotechnol.* **2019**, *37*, 1038–1040.
- (41) Davies, M.; Nowotka, M.; Papadatos, N.; Dedman, N.; Gaulton, A.; Atkinson, F.; Bellis, L.; Overington, J. P. ChEMBL web services: streamlining access to drug discovery data and utilities. *Nucleic Acids Res.* **2015**, *43*, W612–620.
- (42) Wu, H.; Wacker, D.; Mileni, M.; Katritch, V.; Han, G. W.; Vardy, E.; Liu, W.; Thompson, A. A.; Huang, X. P.; Carroll, F. I.; Mascarella, S. W.; Westkaemper, R. B.; Mosier, P. D.; Roth, B. L.; Cherezov, V.; Stevens, R. C. Structure of the human kappa-opioid receptor in complex with JDTC. *Nature* **2012**, *485*, 327–332.
- (43) Irwin, J. J.; Tang, K. G.; Young, J.; Dandarchuluun, C.; Wong, B. R.; Khurelbataar, M.; Moroz, Y. S.; Mayfield, J.; Sayle, R. A. ZINC20-A Free Ultralarge-Scale Chemical Database for Ligand Discovery. *J. Chem. Inf. Model.* **2020**, *60*, 6065–6073.
- (44) Baell, J. B.; Holloway, G. A. New Substructure Filters for Removal of Pan Assay Interference Compounds (PAINS) from Screening Libraries and for Their Exclusion in Bioassays. *J. Med. Chem.* **2010**, *53*, 2719–2740.
- (45) Kalgutkar, A. S.; Gardner, I.; Obach, R. S.; Shaffer, C. L.; Callegari, E.; Henne, K. R.; Mutlib, A. E.; Dalvie, D. K.; Lee, J. S.; Nakai, Y.; O'Donnell, J. P.; Boer, J.; Harriman, S. P. A comprehensive listing of bioactivation pathways of organic functional groups. *Curr. Drug. Metab.* **2005**, *6*, 161–225.
- (46) Kalgutkar, A. S.; Soglia, J. R. Minimising the potential for metabolic activation in drug discovery. *Expert Opin. Drug Metab. Toxicol.* **2005**, *1*, 91–142.
- (47) Bemis, G. W.; Murcko, M. A. The properties of known drugs. 1. Molecular frameworks. *J. Med. Chem.* **1996**, *39*, 2887–2893.
- (48) Polykovskiy, D.; Zhebrak, A.; Sanchez-Lengeling, B.; Golovanov, S.; Tatanov, O.; Belyaev, S.; Kurbanov, R.; Artamonov, A.; Aladinskiy, V.; Veselov, M.; Kadurin, A.; Johansson, S.; Chen, H.; Nikolenko, S.; Aspuru-Guzik, A.; Zhavoronkov, A. Molecular Sets (MOSES): A Benchmarking Platform for Molecular Generation Models. *Front. Pharmacol.* **2020**, *11*, No. 565644.
- (49) Ertl, P.; Rohde, B.; Selzer, P. Fast calculation of molecular polar surface area as a sum of fragment-based contributions and its application to the prediction of drug transport properties. *J. Med. Chem.* **2000**, *43*, 3714–3717.
- (50) RDKit: Open-source cheminformatics; 2010. <https://www.rdkit.org/>.
- (51) Wildman, S. A. C.; G. M. Prediction of Physicochemical Parameters by Atomic Contributions. *J. Chem. Inf. Comput. Sci.* **1999**, *39*, 868–873.
- (52) Delaney, J. S. ESOL: estimating aqueous solubility directly from molecular structure. *J. Chem. Inf. Comput. Sci.* **2004**, *44*, 1000–1005.
- (53) Kingma, D. P. B.; J. A. In *A method for stochastic optimization*, 3rd International Conference for Learning Representations, San Diego, CA, US, 2015.
- (54) OpenEye Python Toolkits. OpenEye, Cadence Molecular Sciences: Santa Fe, NM 2021 <http://www.eyesopen.com>.
- (55) Degen, J.; Wegscheid-Gerlach, C.; Zaliani, A.; Rarey, M. On the art of compiling and using 'drug-like' chemical fragment spaces. *Chem. Med. Chem.* **2008**, *3*, 1503–1507.
- (56) Preuer, K.; Renz, P.; Unterthiner, T.; Hochreiter, S.; Klambauer, G. Frechet ChemNet Distance: A Metric for Generative Models for Molecules in Drug Discovery. *J. Chem. Inf. Model.* **2018**, *58*, 1736–1741.
- (57) Williams, R. J. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Mach. Learn.* **1992**, *8*, 229–256.
- (58) Papadopoulos, K.; Giblin, K. A.; Janet, J. P.; Patronov, A.; Engkvist, O. De novo design with deep generative models based on 3D similarity scoring. *Bioorg. Med. Chem.* **2021**, *44*, No. 116308.
- (59) Quacpac ToolkitOpen. Eye, Cadence Molecular Sciences: Santa Fe, NM, 2021, <http://www.eyesopen.com>.
- (60) Omega Toolkit. OpenEye, Cadence Molecular Sciences: Santa Fe, NM, 2021, <http://www.eyesopen.com>.

- (61) Schreyer, A. M.; Blundell, T. USRCAT: real-time ultrafast shape recognition with pharmacophoric constraints. *J. Cheminf.* **2012**, *4*, 27.
- (62) Tran-Nguyen, V. K.; Da Silva, F.; Bret, G.; Rognan, D. All in One: Cavity Detection, Druggability Estimate, Cavity-Based Pharmacophore Perception, and Virtual Screening. *J. Chem. Inf. Model.* **2019**, *59*, 573–585.
- (63) Da Silva, F.; Desaphy, J.; Rognan, D. IChem: A Versatile Toolkit for Detecting, Comparing, and Predicting Protein-Ligand Interactions. *Chem. Med. Chem.* **2018**, *13*, 507–510.
- (64) Bannan, C. OE_toandfrom_RDK.ipynb. <https://gist.github.com/bannanc/810ccc4636b930a4522636baab1965a6>.
- (65) *LigPrep, Protein Preparation Wizard, QikProp, and Glide*. Schrödinger, LLC New York, NY, 2020.
- (66) Roos, K.; Wu, C.; Damm, W.; Reboul, M.; Stevenson, J. M.; Lu, C.; Dahlgren, M. K.; Mondal, S.; Chen, W.; Wang, L.; Abel, R.; Friesner, R. A.; Harder, E. D. OPLS3e: Extending Force Field Coverage for Drug-Like Small Molecules. *J. Chem. Theory Comput.* **2019**, *15*, 1863–1874.
- (67) Goddard, T. D.; Huang, C. C.; Meng, E. C.; Pettersen, E. F.; Couch, G. S.; Morris, J. H.; Ferrin, T. E. UCSF ChimeraX: Meeting modern challenges in visualization and analysis. *Protein Sci.* **2018**, *27*, 14–25.
- (68) Pettersen, E. F.; Goddard, T. D.; Huang, C. C.; Meng, E. C.; Couch, G. S.; Croll, T. I.; Morris, J. H.; Ferrin, T. E. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Sci.* **2021**, *30*, 70–82.
- (69) Daina, A.; Zoete, V. A BOILED-Egg To Predict Gastrointestinal Absorption and Brain Penetration of Small Molecules. *Chem. Med. Chem.* **2016**, *11*, 1117–1121.
- (70) Jorgensen, W. L.; Duffy, E. M. Prediction of drug solubility from structure. *Adv. Drug Delivery Rev.* **2002**, *54*, 355–366.
- (71) Ertl, P.; Schuffenhauer, A. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *J. Cheminf.* **2009**, *1*, 8.
- (72) You, J.; Liu, B.; Ying, Z.; Pande, V.; Leskovec, J. Graph convolutional policy network for goal-directed molecular graph generation. *Adv. Neurol. Inf. Process. Syst.* **2018**, 6410–6421.
- (73) *GraphPad Prism*; GraphPad Software: San Diego, California, US www.graphpad.com.
- (74) Ballesteros, J. A.; Weinstein, H. Integrated Methods for the Construction of Three-Dimensional Models and Computational Probing of Structure-Function Relations in G Protein-Coupled Receptors. In *Methods in Neurosciences* Elsevier, 1995; Vol. 25, pp 366–428.
- (75) Isberg, V.; de Graaf, C.; Bortolato, A.; Cherezov, V.; Katritch, V.; Marshall, F. H.; Mordalski, S.; Pin, J. P.; Stevens, R. C.; Vriend, G.; Gloriam, D. E. Generic GPCR residue numbers - aligning topology maps while minding the gaps. *Trends Pharmacol. Sci.* **2015**, *36*, 22–31.
- (76) Harding, S. D.; Armstrong, J. F.; Faccenda, E.; Southan, C.; Alexander, S. P. H.; Davenport, A. P.; Pawson, A. J.; Spedding, M.; Davies, J. A.; Ne, I. The IUPHAR/BPS guide to PHARMACOLOGY in 2022: curating pharmacology for COVID-19, malaria and antibacterials. *Nucleic Acids Res.* **2022**, *50*, D1282–D1294.